MIL-L-52841A(ME)
28 May 1981
SUPERSEDING
MIL-L-52841
27 January 1975

MILITARY SPECIFICATION

LABORATORY, FIELD: TRAILER CHASSIS MOUNTED, FOR DEVELOPING PIPELINE WELD RADIOGRAPHS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a laboratory for developing and viewing radiographic film of pipeline welds. The laboratory is portable and is mounted on a trailer chassis.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6635

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-T-416	- Timer, Interval.
SS-P-196	- Pencil: Spiral-Paper-Form, Wax, Glass- and China Marking.
UU-B-625	- Books, Engineers' Field.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.
MIL-G-28554	- Generator Sets, Mobile Electric Power and Supplemental Equipment, Packaging of.
MIL-T-36527	- Thermometer, Self Indicating, Liquid in Glass, Film Processing.
MIL-E-52031	- Extinguisher, Fire Vaporizing-Liquid: CF_3BR; 2-3/4-Pound, with Bracket.
MIL-C-52437	- Chassis, Trailer, Single and Multi-Axle.
MIL-G-52732	- Generator Sets, Gasoline Engine Driven, 0.5 KW Thru 10 KW, 60 Hertz, 400 Hertz and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility) General Specification for.
MIL-A-52767	- Air Conditioners: Vertical and Horizontal, Compact.
MIL-T-52834	- Tank, Master, X-Ray Film Processing, For 70 MM Film.
MIL-V-52837	- Viewer, X-Ray Film: Industrial, Wall-Mounted.
MIL-D-52842	- Drier, X-Ray, Film: Forced Air, Wall-Mounted.

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STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions For Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH).

Threshold Limit Values for Chemical Substances, 1980 Edition.

(Application for copies should be addressed to the American Conference of Governmental Industrial Hygienists, 2205 South Road, Cincinnati, OH 54238).

ALUMINUM ASSOCIATION, THE (AA).

Aluminum Construction Manual, Section 1 - Specification for Aluminum Structures.

(Application for copies should be addressed to the Aluminum Association, 818 Connecticut Avenue, NW, Washington, DC 20006.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM E94 - Radiographic Testing.

(Application for copies should be addressed to the American National Standards Institute, Inc. (ANSI) 1430 Broadway, New York, NY 10018.)

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AMERICAN PETROLEUM INSTITUTE (API)

API 1104 - Standard for Welding Pipeline and Related Facilities.

(Application for copies should be addressed to the American Petroleum Institute, 2101 L Street, NW, Washington, DC 20037.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Nondestructive Examination, Article 22, Radiographic Standards.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The laboratory shall be similar to laboratory assembly 2-1/2 ton trailer mounted (see 6.8) and as specified herein, except that the processing tank (MIL-T-52834), the viewer (MIL-V-52837) and the drier (MIL-D-52842) shall be as specified in the individual specifications and shall be installed by the contractor.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more laboratories for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

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3.5 Environmental requirements.

3.5.1 Operating. The laboratory shall operate as specified herein in any ambient temperature from plus 110° F to minus 25° F.

3.5.2 Storage. The laboratory shall not be damaged by storage under the following conditions:

Air temperature
High - Plus 160° F, 4 hours daily.
Low - Minus 30° F, 12 hours. No solar radiation.

3.6 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair and replacement without the removal of other major assemblies or installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.7 Electrical requirements. The laboratory shall be designed to operate from 120-volt, 60-hertz, single-phase current. All electrical equipment shall be effectively guarded and grounded to prevent electrical shock hazard. Snug-fitting grommets shall be used where wiring passes through holes.

3.7.1 Door switch. A door-operated switch shall be provided so that all lights in the laboratory will be turned off as soon as the door starts to open.

3.7.2 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.7.3), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.7.3 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish preclude the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.8 Construction. The laboratory shall be constructed so that no parts or components will work free in service. When mounted on the Government-furnished trailer chassis, the laboratory shall withstand the strains, jars, vibrations, and other conditions incidental to shipping, transportation, or being towed cross-country and over paved and unpaved roads. The laboratory shall be securely bolted to the trailer chassis in a manner that will permit its removal from the trailer chassis when so required. Electrical connections shall be equipped with waterproof quick-disconnect connections. No modification to the trailer chassis shall be permitted other than removal of minor components or drilling of boltholes for joining the laboratory structure to the trailer chassis. Structural members of the laboratory shall be of sufficient strength to withstand the stresses produced in meeting lifting requirements specified in 3.14 without structural failure.

3.8.1 Weight limitation. The laboratory, without the chassis, shall weigh not more than 5000 pounds when completely loaded with fuel, water, processing chemicals, gamma-ray projector and accessories in their sheets, one day's supply chest filled with necessary supplies, and the Government-furnished weld sample testing equipment. Before finalization of design and fabrication of any components, the contractor shall compute the laboratory's final loaded weight. If at that time the contractor determines that the weight of the complete laboratory is greater than 5000 pounds, he shall notify the contracting officer as soon as possible so that necessary alterations can be made.

3.8.2 Center of gravity. The laboratory body shall be positioned so that its center of gravity will be forward of the trailer chassis wheels so that a downward force of not more than 15 percent nor less than 10 percent of the laboratory's total load (loaded or unloaded) will be exerted by the trailer lunette on the towing vehicle's towing pintle when horizontal. The fully loaded laboratory's center of gravity in the transversal plane shall be on the longitudinal centerline of the laboratory plus or minus one inch. The center of gravity of the loaded laboratory in the transversal plane must move not more than plus or minus 3 inches from the longitudinal centerline of the laboratory due to sloshing of fuel and water during towing of the laboratory over prepared road surfaces. Transverse movement of the center of gravity should favor the left side of the laboratory during towing to correct for normal road surface crown. When unloaded (i.e. no fuel or water), the center of gravity of the laboratory shall be on the longitudinal centerline of the laboratory, plus or minus one inch.

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3.8.3 Insulation. The walls, roof and flooring of the laboratory darkroom shall be insulated. The thickness of the insulation shall be not less than 2 inches, except at framing members. Insulating material shall have a thermal conductivity factor of not less than 0.257 Btu/sq ft/hr/° F/inch thickness and the laboratory shall have an overall thermal conductivity factor of not less than 0.2 Btu/sq ft/hr/° F/inch thickness. The insulating material shall be waterproof, vermin-proof, fungus resistant, nonpacking, and shall not support combustion. The edges of the darkroom door shall be stepped to prevent light leakage and weather-stripped to prevent leakage of heated or cooled air.

3.9 Safety. All rotating, reciprocating or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All exposed parts that are subject to high operating temperatures shall be fully guarded or insulated.

3.10 Noise limits. The noise level of the laboratory at the operator's station and surrounding area shall not exceed the limits specified in MIL-STD-1474, Category D. The precautions of MIL-STD-1474 (see 6.10) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.6.2.6.

3.11 Heating unit. An electric heating unit operating from the generator set shall be provided. The heating unit shall raise the temperature in the laboratory from minus 25° F to plus 40° F within one hour while the outside temperature remains at minus 25° F. An interlock shall be provided so that it will be impossible to turn on the heating unit and the air conditioner at the same time. The heated air from the heating unit shall enter the laboratory near the floor.

3.12 Air conditioner. The air conditioner shall lower the temperature inside the laboratory from plus 110° F to plus 85° F in one hour. The cooled air from the air conditioner shall enter the trailer near the top of the laboratory.

3.13 Blower system. An automatic blower system shall be provided to bring fresh outside air into the laboratory whenever the generator is running. The blower system may be an independent unit or it may be used in conjunction with the air conditioner or heater; however, the blower system shall bring in fresh air whenever the generator is operating whether or not the heater or air conditioner is operating. The incoming air shall be ducted or baffled so that the air will not blow directly on the head or shoulders of the operator. The blower system shall provide ventilation to keep concentrations of gases, vapors, and funes below the American Conference of Governmental Industrial Hygienists' Threshold Limit Values. Intakes for ventilation systems shall be so located as to minimize the introduction of contaminated air.

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3.14 Fire extinguishers. The contractor shall mount one Government-furnished fire extinguisher on the trailer tongue near the entrance door and the other inside the laboratory in any position convenient to personnel inside the laboratory.

3.15 Lifting attachments. The laboratory shall be provided with permanently affixed lifting attachments. The attachments shall enable the laboratory, with trailer chassis attached, to be lifted in its normal operating position without failure or permanent deformation to any part or component of the laboratory. Each attachment shall withstand 2-1/2 times the resulting static force imposed without failure or permanent deformation to any parts of the laboratory when the laboratory is suspended by the lifting attachments. Normal static load includes the weight of the laboratory without fuel and water and the weight of the trailer chassis. Lifting attachments shall conform to MIL-STD-209 for Type II equipment. A clearance not less than one inch shall be maintained between the slings and the laboratory body. Two sets of lifting attachments shall be provided. One set shall be permanently affixed to the top of the laboratory for use in separating the laboratory from the trailer chassis when required. A second set, unless already provided, shall be permanently affixed to the trailer chassis for lifting the complete laboratory. Spreader attachments shall be permanently affixed across the top of the laboratory to maintain the one-inch clearance between the slings and the laboratory sides. The lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.16.

3.16 Tiedown attachments. Permanently affixed attachments that permit fastening down the laboratory to the floor or deck of a transportation medium shall be provided. The attachments shall conform to MIL-STD-209 for Type II equipment. Attachments shall be rounded or beveled to prevent cutting of the tiedown material. Tiedown attachments may be used as lifting attachments when such attachments meet requirements in 3.15.

3.16.1 Interior tiedown attachments. All interior tiedown attachments within the laboratory shall be positioned so as to obtain maximum support from the laboratory's structural members. The attachments shall withstand daily use during off-road transportation over difficult terrain without failure or permanent deformation and shall not cause structural damage to the laboratory.

3.17 Identification marking. The laboratory shall be identified in accordance with MIL-STD-130. The marking shall be applied to the laboratory on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.18 Shipping data plate. The shipping data plate shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. In addition to required markings, the plate shall show the silhouette of the laboratory in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plate shall be attached by screws, bolts or rivets in a conspicuous protected location.

3.19 Instruction plates. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets, in a conspicuous protected location.

3.20 Treatment and painting. The portions of the laboratory normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, Type B, color forest green.

3.20.1 Dissimilar metals. Nonpermissible galvanic couples (dissimilar metals), as defined in MIL-T-704, shall be insulated as specified in MIL-T-704.

3.21 Accessory equipment. The following accessories shall be furnished with each laboratory. All accessories, except the timer, shall be packed in the case provided.

Item	Description	Quantity
1	Penetrameters - No. 5.	6
	ASME Article 22 - No. 6.	6
	Radiographic Standards - No. 8.	6
2	Reels, film processing for 70 mm film.	3 sets
3	Lead letters, 0.50-inch, A through Z.	24 each
4	Lead numbers, 0.50-inch, 0 through 9	24 each
5	Thermometer (see MIL-T-36527).	2
6	Timer (see GG-T-416, Type I, Class 2), permanently mounted in a position over the processing tank with its dial illuminated by the safelight.	1
7	Mixing paddle or stick, corrosion-resisting steel.	1
8	Scissors, common office type, 9-inch.	2
9	Tape, measuring, 6-foot, pull-push, flexible.	1
10	Stoppers, rubber, for developing tank.	4

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3.22 Operating supplies. The following expendable supplies shall be furnished with each laboratory. All supplies shall be new and shall be packed in the case provided.

Item	Description	Quantity
1	Film, 70 mm, lead-pack, industrial, X-ray.	3200 feet
2	Tape, friction or masking, 1 inch x 60 yard adhesive backing.	2 rolls
3	Film developing chemical, dry powder, packaged.	2 five-gallon batches
4	Film fixing chemical, dry powder, packaged.	2 five-gallon batches
5	Film developer refresher chemical: for 1 gallon.	1
	for 5 gallons.	1
6	Notebook UU-B-625, Type I, Ruling C.	2
7	Pencils, common, No. 2 Grade.	12
8	Pencil sharpener, hand-held type (not wall-mounted type).	1
9	Crayon, spiral-paper-form, yellow, SS-P-196.	10
10	Towel, paper, 11-inch roll.	4
11	Sponge, synthetic, 4 inches x 6 inches.	6
12	Wiping rags, bundles 2 pounds each.	4
13	Filter cartridges for washwater.	4
14	Grease pencils, black, SS-P-196.	4
15	Copy of API 1104.	1
16	Copy of ANSI/ASTM E94	1

3.23 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.4):

Item No.	Description	Identification	Qty. for each laboratory
1	Chassis, Trailer: Trailing Arm, 4-Wheel, 2-1/2 Ton.	MIL-C-52437	1
2	Generator set, gasoline-engine driven, 5.0 KW, 60 Hz, alternating current, air-cooled, portable, tubular frame, skid mounted.	MIL-G-52732, Type I; NSN 6115-00-017-8240	1

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Item No.	Description	Identification	Qty. for each laboratory
3	Air conditioner, vertical compact, 6000 Btu per hour.	MIL-A-52767, Type I, Class 1; NSN 4120-00-926-1139	1
4	Extinguisher, fire, monobromo-trifluoromethane.	MIL-E-52031, NSN 4210-00-555-8837	2
5	Viewer, for 70 mm X-ray film, wall-mounted.	MIL-V-52837	1
6	Drier, X-ray film: forced-air, wall-mounted.	MIL-D-52842	1
7	Tank, X-ray film processing, for 70 mm film.	MIL-T-52834	1

3.24 Transportability.

3.24.1 Rail. Unless otherwise specified (see 6.2), the unpackaged laboratory shall withstand rail humping without damage or permanent deformation to any part or component when tested as specified in 4.6.2.9.1. After testing, the laboratory shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the laboratory.

3.25 Workmanship. The laboratory shall be free from defects such as misalined components, incomplete welds, cracks, burns, leaks, and other defects that could impair the operation and serviceability of the laboratory. All parts shall be clean and free from dirt, sand, grease, oil, and metal chips.

3.25.1 Castings and forgings. All parts, components and assemblies of the laboratory which include castings and forgings shall be cleaned of harmful foreign material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.25.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that methods that will not cause damage to the material. Corners shall be square and true. Shearing and sawing shall be done neatly and accurately. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. Surfaces shall be free from burrs, sharp edges and corners.

3.25.3 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers and lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

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3.25.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member. Rivets and riveting shall be in accordance with SAE J429.

3.25.5 Welding (aluminum). Aluminum welding shall be in accordance with 7.2 of AA Aluminum Construction Manual, Section 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article laboratory shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for

4.3.2 Tests. The first article laboratory shall be tested as specified in 4.6.2.1 through 4.6.2.9.1. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production laboratories will be selected at random by the Government from the laboratories being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.9.1 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production laboratory shall not exclude the remaining laboratories from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production laboratory to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production laboratory and shall be cause for refusal by the Government to continue acceptance of production laboratories until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on laboratories previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all laboratories accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each laboratory shall be examined as specified in 4.6.1. Evidence of any defect shall be cause for rejection.

4.5.2 Tests. Each laboratory shall be tested as specified in 4.6.2.1, except that the towing distance shall be not less than 5 miles for paved roads, 5 miles for unpaved roads, and 2 miles cross-country, and the test conducted only one time. Any evidence of damage to any part of the laboratory or its contents shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The laboratory shall be examined as specified herein for the following defects:

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101. Materials not as specified.
102. Safety devices or guards missing or not adequate.
103. Electrical components not properly grounded.
104. Door switch does not turn off all lights when it starts to open.
105. Blower does not automatically start and run when the generator set is running or does not provide ventilation as specified.
106. Any component such as, but not limited to, light switches, lamps, motor, or safelight does not function as specified.
107. Timer not mounted permanently over the processing tank and in full view of the safelight.
108. Accessory and operating components not as specified or not packed in the chests.
109. Fire extinguishers not mounted as specified.
110. Identification markings incorrect, illegible or missing.
111. Shipping data plate missing or not as specified.
112. Treatment and painting not as specified.
113. Workmanship not as specified.

4.6.2 Tests. Prior to any testing, the trailer chassis and the generator set shall be serviced and lubricated in accordance with their respective instructions.

4.6.2.1 Road test. With the water tank filled and all accessory and operating equipment properly stored in the appropriate chests, the laboratory, mounted on the trailer chassis, shall be towed as follows:

(a) Fifty miles over paved roads at maximum speed up to 45 miles per hour (mph). Not less than 25 miles shall be at a speed of 45 mph.
(b) Fifty miles over dirt roads at maximum speeds up to 25 mph. Not less than 25 miles shall be at a speed of 25 mph.
(c) Twenty-five miles over cross-country terrain. The terrain shall be sufficiently rough to approximate conditions encountered in pipeline construction. The vehicle shall cross small gullies, large stones and similar obstacles.

This test shall be repeated 10 times for a total distance of 1250 miles. Any evidence of damage to the laboratory or any installed component or breaking loose of any chest shall constitute failure of this test.

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4.6.2.2 Cold-temperature storage test. Prior to the cold-temperature storage test, all water shall be drained from the laboratory. The laboratory shall be placed in the test chamber and the temperature in the test chamber lowered to minus 30° F. This temperature shall be maintained for 12 hours.

4.6.2.3 Cold-temperature operational test. At the completion of the cold-temperature storage test the temperature in the test chamber shall be raised to minus 25° F and allowed to stabilize. The heater shall be operated by the generator in the laboratory. Any damage caused by the storage test, failure of the heater to raise the temperature inside the laboratory to plus 40° F in one hour, or failure of any component to operate as specified herein shall constitute failure of this test.

4.6.2.4 Hot storage test. The temperature in the test chamber shall be raised to plus 160° F and maintained for 4 hours. The generator shall not be in operation during this test.

4.6.2.5 Warm operational test. The temperature in the test chamber shall be lowered to plus 110° F and allowed to stabilize. The air conditioner shall then be operated by the generator in the laboratory. Any damage caused by the hot storage test, failure of the air conditioner to lower the temperature inside the laboratory to plus 85° F in one hour, or failure of any component to operate as specified herein shall constitute failure of this test.

4.6.2.6 Noise level measurement. Measure the noise level of the operating station and surrounding area, while operating the laboratory at the maximum noise condition. Should the noise level exceed 85 dB(A) at this condition, failure to provide the precautions specified in 3.10 shall constitute failure of this test.

4.6.2.7 Lifting attachment test. The laboratory, without the trailer chassis, shall be firmly bolted to the floor of the test site. The lifting sling shall be attached to the lifting eyes on the top of the laboratory and an upward pull of 2-1/2 times the weight of the laboratory exerted on the lifting eyes. Any evidence of damage or permanent deformation of the laboratory or lifting eyes or breaking of welds shall constitute failure of this test.

4.6.2.8 Tiedown attachments. The tiedown attachments shall be securely attached to a platform and subjected to the test specified in 4.6.2.7. Nonconformance to 3.16 shall constitute failure of this test.

4.6.2.8.1 Interior tiedown attachments. After the road test the interior tiedown attachments should be examined. Nonconformance to 3.16.1 shall constitute failure of this test.

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4.6.2.9 Transportability.

4.6.2.9.1 Railroad car impact. Anchor the unpackaged laboratory on a domestic service railway freight car (blocking is permitted). Hump the car, with the laboratory tied down, at speeds of not less than 4, 6, and 8 mph respectively against 5 empty freight cars with their brakes set and draft gear extended. The laboratory shall face in the direction of the 5 cars. Repeat the procedure at not less than 8 mph with the rear of the laboratory in the direction of the 5 cars. The laboratory may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the laboratory in the proper direction of the freight car. Once the test has begun, there shall be no readjustment of load, nor any reconditioning of the blocks, chocking material, or tiedowns except as necessary to reverse the direction of the laboratory. After each impact, the laboratory shall be inspected. After completion of the railroad impact test, an operational checkout of the laboratory shall be conducted. Nonconformance to 3.24.1 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select laboratories at any time during the contract production period and subject these laboratories to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on laboratories selected at random from those which have been accepted by the Government and will not include the previously inspected first article model and initial production laboratories. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection camparison laboratory to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison laboratory and shall be cause for refusal by the Government to continue acceptance of production laboratories until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on laboratories previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all laboratories accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection. The preproduction pack shall be examined for the defects specified in 4.8.2.4. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Inspection stages. Inspection shall be in three stages as follows:

(a) The first stage shall include inspection of procedures, methods, materials, containers, and markings prior to placing items into consolidated containers.
(b) The second stage shall include inspection of closure and marking of consolidated containers.
(c) The third stage shall include inspection of the arrangement, blocking, bracing, cushioning, and anchoring of boxed items within the laboratory inclosure and preparation and marking of the laboratory prepared for shipment.

4.8.2.2 Unit of product. For the purpose of inspection, a completed pack as prepared for any inspection stage shall be considered a unit of product.

4.8.2.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.4 Examination. Samples selected in accordance with 4.8.2.3 shall be examined for the defects marked "X" for the applicable level in Table I. AQL shall be 2.5 percent defective.

Table I. Classification of defects

Defect	Level	
	A	Commercial
114. Materials, methods, procedures, or containers not as specified. Each incorrect material, method, procedure, or container shall constitute one defect.	X	-
115. Fire extinguishers not removed and boxed as specified.	X	-
116. Film reels not placed and secured as specified.	X	-
117. Lead letters not boxed as specified.	X	-
118. Lead numbers not boxed as specified.	X	-
119. Filter cartridges not boxed as specified.	X	-
120. Louvers of field laboratory not covered as specified.	X	-
121. Consolidation not as specified.	X	-
122. Preserved and consolidated components not immobilized within the laboratory inclosure as specified.	X	-
123. Components not preserved in accordance with referenced documents as specified.	X	-
124. Preservation and packing not in accordance with the referenced document as specified.	-	X
125. Marking illegible, incomplete, incorrect, or missing.	X	X

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5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be prepared utilizing either the preproduction model or a production laboratory. When the preproduction model laboratory is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production laboratories.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Methods of preservation. Methods of preservation specified herein shall conform to the applicable methods listed in MIL-P-116.

5.2.1.2 Trailer chassis. Trailer chassis components shall be preserved in accordance with MIL-STD-281, Level A, as specified for mobile vehicles.

5.2.1.3 Generator set. The generator set shall be preserved in accordance with MIL-G-28554, Level B.

5.2.1.4 Fire extinguishers. Each fire extinguisher shall be removed from the bracket and placed in a box conforming to PPP-B-636, Type CF, Variety SW, Grade 125, style optional or in the contractor's carton.

5.2.1.5 Film reels. Film reels shall be placed and secured in the racks and the rack shall be secured in the holders provided in the laboratory inclosure.

5.2.1.6 Lead letters. Lead letter sets shall be placed together in a close-fitting box conforming to PPP-B-636, Type CF, Variety SW, Grade 125, style optional or in the contractor's carton.

5.2.1.7 Lead numbers. Lead number sets shall be placed together in a close-fitting box conforming to PPP-B-636, Type CF, Variety SW, Grade 125, style optional.

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5.2.1.8 Thermometers and pencil sharpeners. Each item shall be individually preserved Method IC-1 or IC-3.

5.2.1.9 Stoppers, masking tape, notebooks, pencils, crayons, towels, sponges, and wiping rags. Items of like description shall be preserved together Method IC-1 or IC-3.

5.2.1.10 Filter cartridges. Filter cartridges shall be placed together in a close-fitting box conforming to PPP-B-636, Type CF, Variety SW, Grade 125, style optional or in the contractor's carton.

5.2.1.11 Scissors and measuring tape. The scissors and measuring tape shall be preserved in accordance with PPP-P-40, Level A.

5.2.1.12 Film developing, fixing, and refresher chemicals. Film developing, fixing and refresher chemicals in the contractor's container shall not require additional preservation.

5.2.1.13 Repair parts. The preservation application criteria and applicable submethods of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.14 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

5.2.1.15 Technical publications. Technical publications shall be preserved Method IC-1 or IC-3.

5.2.1.16 Laboratory inclosure. All louvers of the laboratory inclosure and air conditioner opening to the outside air shall be covered with barrier material conforming to PPP-B-1055, Class L4. The barrier shall be secured with tape conforming to MIL-T-22085, Type II.

5.2.1.17 Penetrameters in case, and mixing paddle. These items shall not require preservation.

5.2.1.18 Consolidation. The thermometer, pencil sharpeners, stoppers, masking tape, notebooks, pencils, crayons, towels, sponges, wiping rags, scissors, measuring tape, technical publications, repair parts, maintenance tools, and chemicals shall be consolidated together in the minimum number of boxes conforming to PPP-B-636, W5c, style optional. Cushioning to prevent movement or damage of contents shall be provided. The size and configuration of the boxes shall be determined by the available space that will permit placement and securing of the boxes and all other components within the laboratory inclosure.

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5.2.2 Commercial. Each laboratory with required components shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A. The laboratory shall be prepared for unboxed mobile shipment. Components shall be packed in any available space within the laboratory inclosure. Unboxed items shall be cushioned and blocked to prevent damage to the items. Boxes shall be immobilized within the laboratory inclosure using blocking, bracing, cushioning, and anchoring as required, applied in a manner not to damage the items or components to which the boxes are secured. All doors of the laboratory inclosure shall be closed and secured in a manner to prevent accidental opening or pilferage of the contents.

5.3.2 Commercial. Each complete laboratory preserved as specified in 5.2, shall be packed for unboxed mobile shipment in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military packaging shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The laboratory is intended for use by troops in remote areas, under widely varying environmental conditions, for the development of radiographs of circumferential or girth welds for 8-inch through 24-inch petroleum pipelines. The intent of the radiography is to determine extent of defects, if any, in the pipeline welds or in other type weldments as applicable.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is required for inspection and approval (see 3.2, 4.3 and 6.3).
(c) When initial production inspection is required and number of laboratories to be furnished (see 3.3).

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(d) When the Government-furnished property is required (see 3.23).
(e) When rail humping test is not required (see 3.25.1).
(f) Frame required for submission of the preproduction pack (see 5.1).
(g) Degree of preservation and packing required (see 5.2 and 5.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should consist of one or more units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Government-furnished property. When specified, the contracting officer should arrange to furnish the property specified in 3.23.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the laboratory and components in accordance with this specification.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manuals to be furnished with each laboratory.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary, including any special tools, and instructions on shipment of laboratories.

6.8 Information drawings. Top Assembly TA13220E5100 shows a type of laboratory assembly which has been found acceptable; however these drawings are for information purposes only and are not intended to preclude the furnishing of another laboratory which conforms to this specification. When desired, the contractor should request copies of the drawings from the contracting officer.

6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

6.10 Noise limits. When the noise level of the laboratory exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the laboratory, the distance from the laboratory at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

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